

6 (long) Fig. 6a shows a simplified lengthwise view of an oil tanker with two identical storage tanks, A and B. The empty ship weighs 200,000 tonnes and each tank has capacity for 160,000 tonnes of oil. The tanker is made of steel plates of a single thickness, with Young's Modulus 210 GPa and yield stress 350 MPa.

When tank A is full and tank B is empty, and the ship is floating in still water, the distributed reaction force applied by the sea to the ship has the form shown in Fig. 6b.

(a) Find the values of R_L and R_R defined in Fig. 6b, assuming the weight of the ship is uniformly distributed along its length, and using a value for g of 10 m s^{-2} . [12]

(b) For this loading, derive formulae for the shear force and bending moment distributions along the length of the ship **within tank A only**. Sketch these distributions, marking salient points. [10]

(c) If the ship has a uniform cross-section as shown in Fig. 6c, estimate the minimum thickness of steel plate required so that the steel remains elastic under this loading. Briefly state and discuss any assumptions you make. [8]

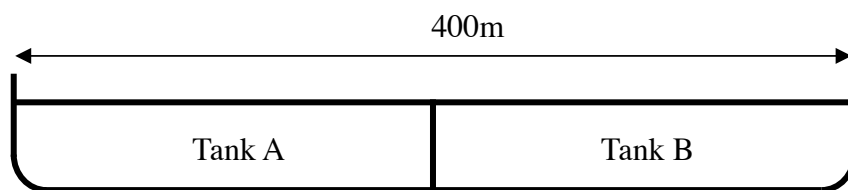


Fig. 6a

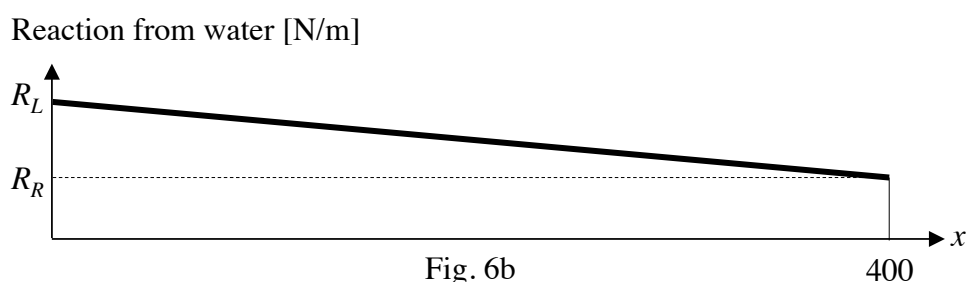


Fig. 6b

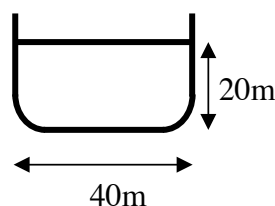


Fig. 6c

NOT TO SCALE